Computer Science and Engineering
National Institute of Technology Rourkela

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Multimodal Deep Learning for Lung Cancer Detection and Characterization

Phase 1 Report: CT-only Baseline for Lung Nodule Classification on LUNA16

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Abstract

This report describes the first phase of a project on multimodal deep learning for lung cancer. Phase 1 builds a CT-only reference model on the LUNA16 benchmark, where each candidate location found by an automatic detector must be classified as a nodule or not a nodule. Each candidate is turned into a 50 mm, three-view (axial, coronal, sagittal) patch, and four ImageNet-pretrained CNNs (ResNet50, DenseNet121, EfficientNet-B0 and VGG16-BN) are fine-tuned on a patient-level train/validation/test split. On the held-out test scans the four models reach ROC-AUC values of 0.686–0.709, and a soft-voting ensemble of all four gives the best result (ROC-AUC 0.720, PR-AUC 0.484 against a chance level of 0.25, F1 0.501). Calibration and Grad-CAM heat maps are also reported. These results are the CT baseline that the PET/CT fusion models of Phase 2 and the radiomics models of Phase 3 will be compared against.

1 What Phase 1 is for

The final goal of this project is a model that combines several kinds of lung-cancer data (CT, PET, clinical and radiomic features). Before mixing modalities we need to know how well a model can do with CT alone. Without that number we cannot say whether fusion actually helps. Phase 1 builds this CT-only reference point.

The task we chose is the *false-positive reduction* step of the LUNA16 challenge [2]. An automatic detector has already marked many suspicious points in each scan ("candidates"). Most of them are vessels, airway walls or other normal tissue. Our model looks at a small patch around each candidate and decides:

- label 1: the candidate is a real nodule (one that radiologists agreed on), or
- label 0: the candidate is not a nodule.

This label says *nodule or not*. It does *not* say benign or malignant. LUNA16 does not provide malignancy labels, so we do not claim to detect cancer in this phase. Histology (cancer subtype) is handled in Phase 2 with the LUNG-PET-CT-Dx data.

We chose this task for three reasons. It is a standard, well-defined benchmark with published results, so we can compare against other work. The labels come from four experienced radiologists. And the data are clean enough that we can test the full pipeline (preprocessing, splitting, training, evaluation, explainability) before moving to the harder multimodal data. Figure 1 shows the whole Phase 1 pipeline.

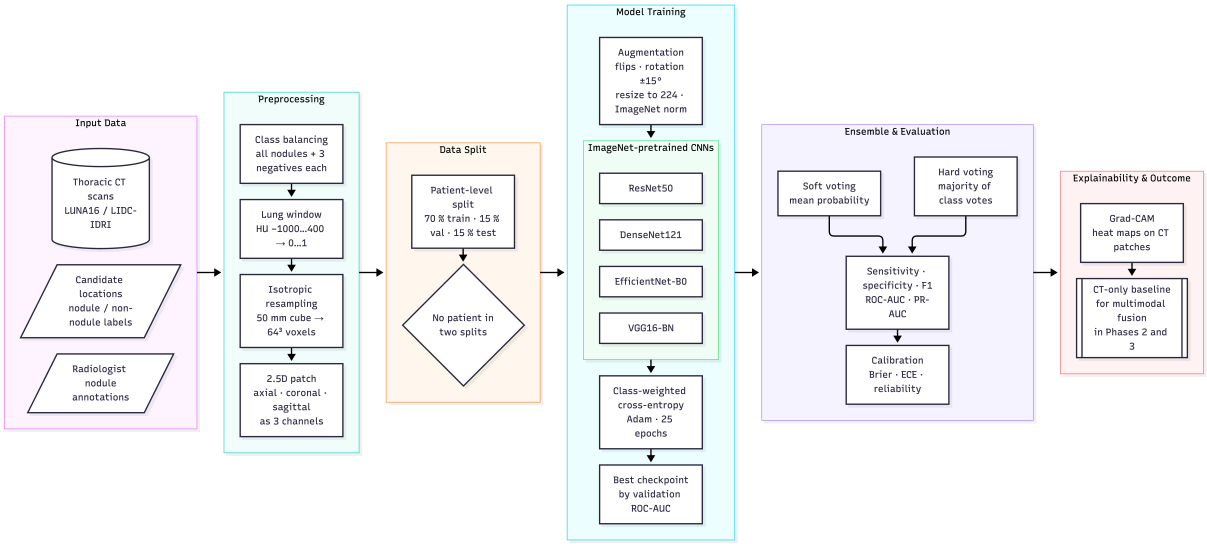


Figure 1: Phase 1 architecture: input data, preprocessing, patient-level split, model training, ensembling and evaluation, and explainability.

2 Dataset

2.1 LIDC-IDRI and LUNA16

LIDC-IDRI [1] contains 1,018 thoracic CT scans from 1,010 patients. Each scan was read by four radiologists in two rounds, and each reader marked nodules of 3 mm or larger. LUNA16 [2] is a cleaned subset of LIDC-IDRI. It keeps 888 scans with slice thickness of at most 2.5 mm and drops scans with missing slices or inconsistent spacing. A nodule counts as a reference nodule if it is at least 3 mm and at least three of the four radiologists marked it. This gives 1,186 nodules.

The scans are already split into ten subsets, and each scan is stored as a `.mhd` header with a `.raw` volume.

We used two of the CSV files that come with LUNA16:

- `candidates_V2.csv` (or `candidates.csv` if V2 is missing): one row per candidate, with the scan ID, the candidate position in millimetres, and the 0/1 label. The organisers recommend the V2 file because the V1 list missed some nodules.
- `annotations.csv`: the centre and diameter of each reference nodule. We only use it to show the nodule size distribution in the EDA.

2.2 Class imbalance and sampling

Well under one percent of the candidates are real nodules. Training on that ratio would teach the model to answer "not a nodule" almost every time. We kept every positive candidate and randomly sampled three negatives per positive (seed 42). This 1:3 ratio still reflects that negatives are more common, and the dataset stays small enough to train four networks on a Kaggle GPU. We keep only candidates whose scan is actually present in the Kaggle copy, so the ratio is correct even if some subsets are missing. Final counts: [N positives], [N negatives], from [N scans].

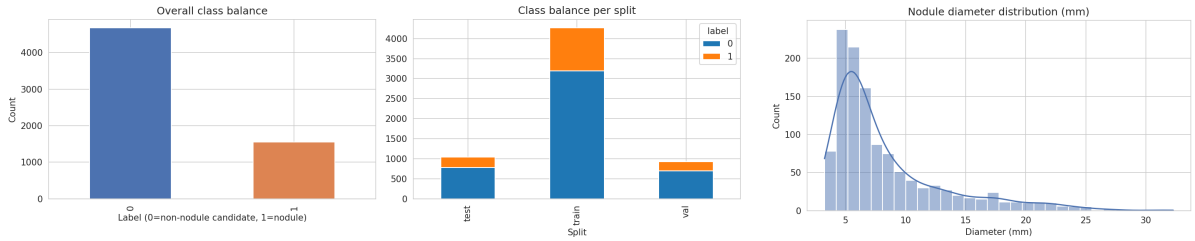


Figure 2: Left: class balance after sampling, overall and per split. Right: diameter of the reference nodules (from `annotations.csv`) in the scans we used. Most nodules are small, which is why a fixed 50 mm patch is enough.

3 Preprocessing

3.1 Intensity window

CT values are in Hounsfield units (HU). Air is about -1000 HU, lung tissue is around -800 to -500, soft tissue and nodules are near 0–100, and bone is several hundred and above. We clip every scan to $[-1000, 400]$ HU and scale it linearly to $[0, 1]$. This is the same window used by Setio et al. [3]. It keeps everything we need to see a nodule against the lung, and it removes very bright bone and metal values that would otherwise take up most of the intensity range.

3.2 Fixed physical size

LUNA16 scans come from different scanners. In-plane pixel size ranges from roughly 0.5 to 1 mm, and slice thickness ranges up to 2.5 mm. If we simply cut 64×64 pixels around each candidate, a patch from one scan would cover a much larger area than a patch from another scan, and the same nodule would look bigger or smaller depending on the scanner. To avoid this, we resample a cube of $50 \times 50 \times 50$ mm around each candidate onto a regular grid of $64 \times 64 \times 64$ voxels (about 0.78 mm per voxel in every direction). The 50 mm size follows Setio et al. [3]. It is large enough to hold the biggest LUNA16 nodules and still shows some surrounding tissue. Candidate positions are converted from millimetres to voxels with SimpleITK, which uses the

full scan geometry (origin, spacing and orientation). Points that fall outside the scan are filled with air.

3.3 2.5D input: three orthogonal views

From each cube we take the three central planes: axial, coronal and sagittal. We stack them as the three channels of one image. This is the "2.5D" idea of Roth et al. [4], and it is close to the multi-view approach of Setio et al. [3].

The reason is simple. In a single axial slice, a nodule and a blood vessel seen end-on both look like a small bright circle. The difference shows up in the other directions: a vessel continues as a tube, while a nodule stays compact. Three views give the network this 3D information. At the same time, the input is still a normal 3-channel 2D image, so we can use networks pretrained on ImageNet. Copying one axial slice into all three channels, which is a common shortcut, would give the network no extra information.

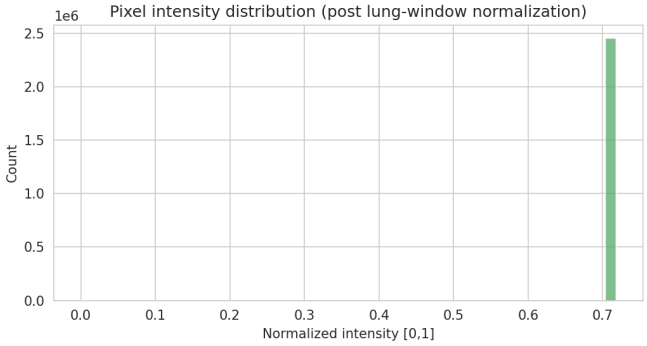


Figure 3: Pixel values after the lung window, from 200 random patches. Expected shape: a large peak near 0 (air in the lung) and a smaller mass to the right (soft tissue and nodules), with nothing outside $[0, 1]$.

3.4 Patient-level split

We split by scan, not by patch: 70% of scans go to training, 15% to validation and 15% to testing. All candidates from one scan go to the same split, and the code checks that no scan appears in two splits. This matters because candidates from the same patient are strongly correlated, and several positive candidates often lie on the same nodule. If patches from one patient appeared in both training and testing, the model could recognise the patient instead of learning what a nodule looks like, and the test score would be too high. Yagis et al. [5] showed this kind of leakage inflating results in 2D medical image classification. Patients per split: [train / val / test].

4 Models

4.1 Transfer learning

A few thousand patches are not enough to train a deep CNN from scratch. Shin et al. [6] and Tajbakhsh et al. [7] found that starting from ImageNet weights [8] and fine-tuning the whole network works as well as or better than training from scratch on medical images, and it converges faster. We therefore start every model from ImageNet weights and replace only the final layer with a 2-class output.

ImageNet models expect 224×224 images normalised with the ImageNet mean and standard deviation. Our patches are 64×64. If we fed them in directly, the last convolutional layer of

ResNet50 or VGG16 would be only 2×2, which leaves almost no spatial detail. So we upsample each patch to 224×224 and apply the ImageNet normalisation, so the input matches what the pretrained weights were trained on.

4.2 The four backbones

We compare four well-known CNN families. Each one is built differently, which helps us understand what matters for this task, and which is also useful for the ensemble.

- **ResNet50** [9]. Skip connections add the input of each block to its output, so very deep networks can still train. It is the most common baseline in medical imaging.
- **DenseNet121** [10]. Each layer receives the outputs of all earlier layers in its block. Early edge and texture features stay available to the last layers, which can help with small objects such as nodules.
- **EfficientNet-B0** [11]. A small network (about 5.3 million parameters) built from mobile inverted-residual blocks with squeeze-and-excitation. It tells us whether a light model can match the larger ones.
- **VGG16-BN** [12, 13]. A plain stack of 3×3 convolutions with batch normalisation and no skip connections. It is an older design, included as a simple reference.

4.3 Training setup

All four models use the same settings, so the comparison is fair (Table 1).

Table 1: Training settings shared by all Phase 1 models.

Setting	Value
Input	3 orthogonal views, 64×64 upsampled to 224×224, ImageNet normalisation
Augmentation (train only)	horizontal and vertical flip ($p = 0.5$), rotation $\pm 15^\circ$
Loss	cross-entropy with class weights (inverse class frequency in training set)
Optimiser	Adam [14], learning rate 10^{-4}
LR schedule	reduce on plateau of validation ROC-AUC (patience 3)
Batch size / epochs	32 / 25
Model selection	checkpoint with the best validation ROC-AUC
Seed	42

Flips and small rotations are safe for nodules because a nodule has no fixed orientation. Class weights make a missed nodule cost about three times as much as a false alarm, which roughly undoes the 1:3 sampling. This keeps sensitivity reasonable at the default threshold of 0.5. For screening, missing a nodule is worse than a false alarm. The side effect is that the predicted probabilities are pushed slightly upwards, so we also measure calibration (below).

4.4 Voting ensembles

Combining several different models usually beats each model alone, as long as the models make different mistakes [15, 16]. Our four backbones are built quite differently, so their errors should not all overlap. We test two simple ensembles:

- **Hard voting**: each model votes 0 or 1. The candidate is called a nodule if at least half the models say so. A 2–2 tie counts as a nodule, again to favour sensitivity.

- **Soft voting**: we average the four predicted nodule probabilities and use a threshold of 0.5.

The ensembles have no trainable weights, so they cannot overfit the test set.

5 Evaluation

All numbers are on the held-out test scans, which are used only once at the end.

- **Sensitivity (recall)** and **specificity**: the share of real nodules we catch, and the share of non-nodules we correctly reject. Clinicians usually ask for these two.
- **Precision** and **F1**: how many flagged candidates are real nodules, and the balance between precision and recall.
- **ROC-AUC**: how well the model ranks nodules above non-nodules, across all thresholds.
- **PR-AUC** (average precision): more informative than ROC-AUC when positives are the minority class [17]. A random model scores about 0.25 here (the positive share), not 0.5.
- **Brier score** [18] and **expected calibration error (ECE)** [19]: whether a predicted probability of, say, 0.8 really means that about 80% of such candidates are nodules. Modern deep networks are often over-confident [19], so we check this and plot reliability diagrams.

Accuracy is also reported. It is not our main metric, though, because with a 1:3 ratio a model that always says "not a nodule" already gets 75%.

5.1 Explainability

We use Grad-CAM [20] on the last convolutional block of each model. It produces a heat map showing which parts of the patch pushed the model towards its prediction. We show it on the axial view for three correctly and three wrongly classified test patches per model. A good model should focus on the nodule itself. If the heat map sits on the patch border or on unrelated tissue, the model may be using a shortcut. The maps are in Figure 9.

6 Results

Table 2 gives the test-set results. The best value in each column is in bold. For Brier score and ECE, lower is better.

Table 2: Test-set results at threshold 0.5.

Model	Acc.	Sens.	Spec.	Prec.	F1	ROC-AUC	PR-AUC	Brier	ECE
ResNet50	0.683	0.496	**0.746**	0.395	0.440	0.686	0.440	0.210	0.176
DenseNet121	0.675	0.581	0.706	0.397	0.472	0.706	0.447	0.206	**0.173**
EfficientNet-B0	0.550	**0.800**	0.466	0.333	0.471	0.699	0.441	0.223	0.222
VGG16-BN	0.666	0.608	0.686	0.392	0.477	0.709	0.460	0.215	0.197
Hard voting	0.649	0.642	0.651	0.380	0.478	0.695	0.391	0.258	0.227*
Soft voting	**0.697**	0.608	0.727	**0.426**	**0.501**	**0.720**	**0.484**	**0.202**	0.179

*The hard-vote "probability" is the share of models voting positive, so it can only take five values. Its Brier score and ECE are not comparable with the others.

6.1 Training behaviour

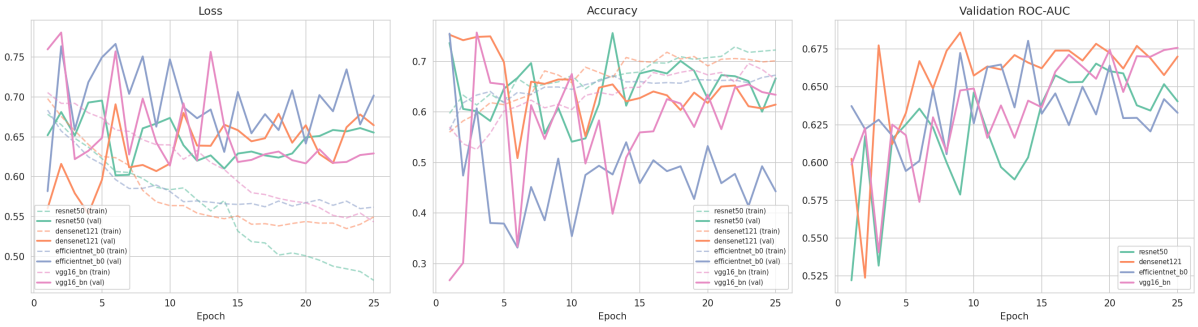


Figure 4: Loss, accuracy and validation ROC-AUC per epoch for all four models (dashed = train, solid = validation). A growing gap between the two curves means overfitting; the saved checkpoint is the epoch with the best validation ROC-AUC.

6.2 Comparison of models and ensembles

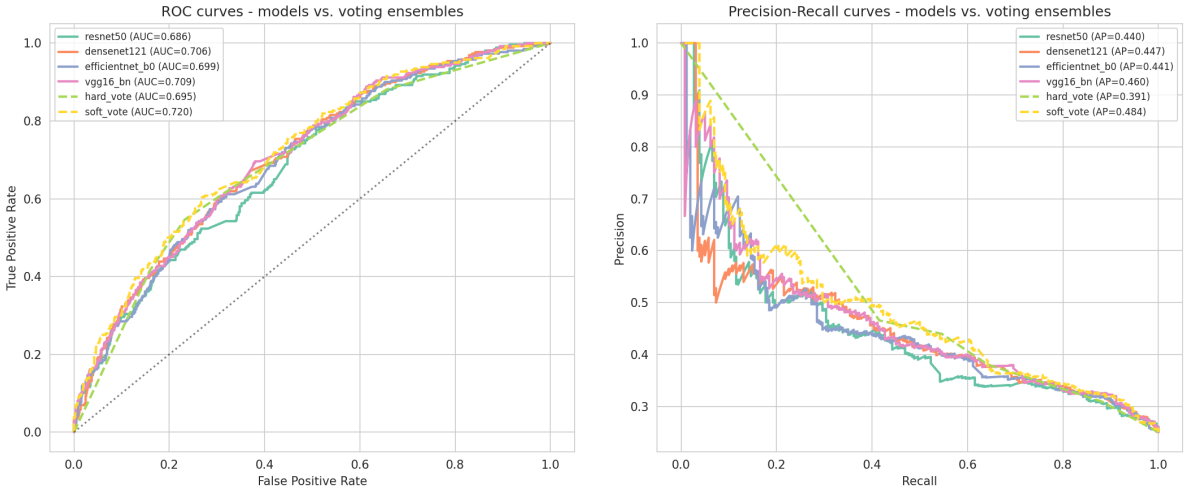


Figure 5: Test-set ROC curves (left) and precision–recall curves (right). Dashed lines are the voting ensembles.

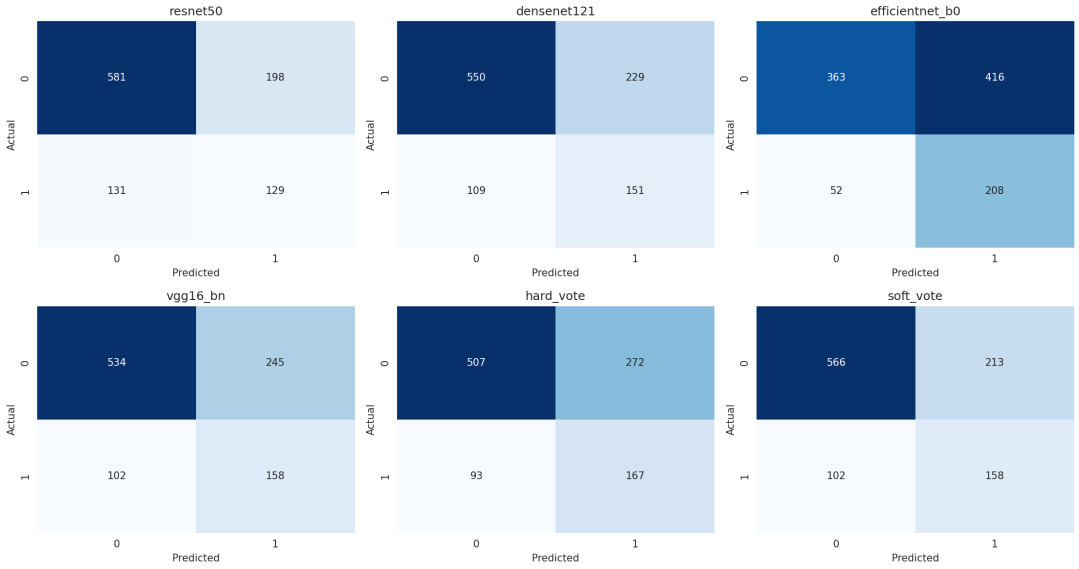


Figure 6: Confusion matrices on the test set at threshold 0.5 (rows = true class, columns = predicted class).

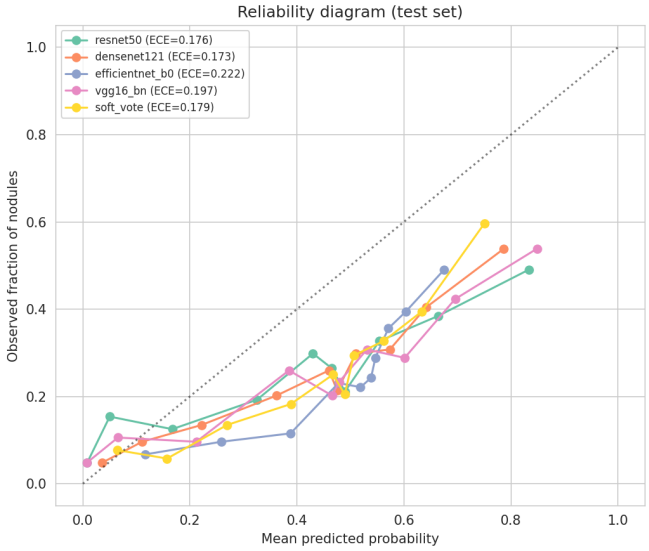


Figure 7: Reliability diagram. Points on the dotted diagonal mean the predicted probability matches how often the candidate is really a nodule. Points below the diagonal mean the model is over-confident.

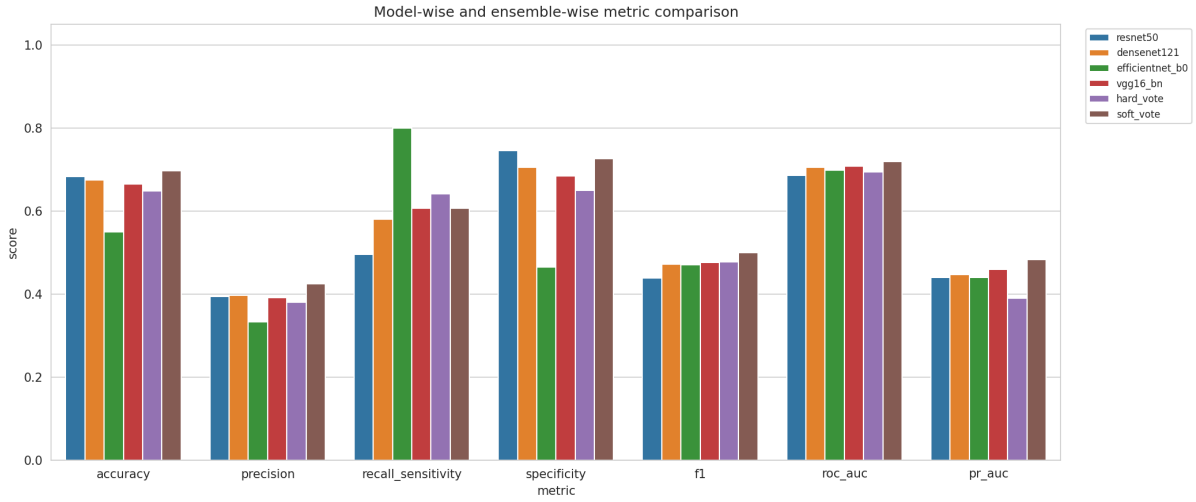


Figure 8: All metrics side by side for each model and ensemble.

6.3 Discussion

The four backbones perform about the same. ROC-AUC lies between 0.686 (ResNet50) and 0.709 (VGG16-BN), and PR-AUC between 0.440 and 0.460. VGG16-BN is the best single model on both ranking metrics and DenseNet121 is a close second. We did not compute confidence intervals for this single test split. With gaps of about 0.02, though, we cannot claim that one architecture is really better than another. Depth, skip connections and dense connections make little difference for this input.

Soft voting is the best model overall. Averaging the four probabilities gives the highest ROC-AUC (0.720), PR-AUC (0.484), F1 (0.501), accuracy (0.697), precision (0.426) and the lowest Brier score (0.202). It improves on the best single model by about 0.01 ROC-AUC and 0.02 PR-AUC. This fits the idea that the four networks make partly different mistakes [16], but the gain is small. Hard voting is worse than soft voting on every ranking metric (ROC-AUC 0.695, PR-AUC 0.391). The reason is that a 0/1 vote throws away how sure each model is, and four votes give only five possible scores.

The models use 0.5 differently. At the same threshold, EfficientNet-B0 flags many candidates (sensitivity 0.80, specificity 0.47), while ResNet50 is much stricter (sensitivity 0.50, specificity 0.75). Their ROC-AUCs are close, so the difference is mostly where their probabilities fall, not how well they separate the classes. For this reason we compare models mainly on ROC-AUC and PR-AUC. For screening, the threshold should be chosen on the validation set to reach a target sensitivity, rather than fixed at 0.5.

Calibration. ECE lies between 0.17 and 0.22 and the Brier score between 0.20 and 0.22. The soft-voting ensemble has the lowest Brier score, and DenseNet121 the lowest ECE. The reliability diagram (Figure 7) shows that predicted probabilities sit above the observed nodule rate. This is the expected effect of the class-weighted loss, which pushes probabilities upwards to protect sensitivity (see the training setup). If the outputs are to be read as nodule risks, a simple recalibration step such as temperature scaling on the validation set [19] can be added without retraining.

Relation to published work and next steps. All models score well above chance on PR-AUC (0.44–0.48 against 0.25), so the three-view patches carry useful information about whether a candidate is a nodule. Our numbers are candidate-level results on one patient-level split. The LUNA16 challenge instead reports FROC and CPM scores from 10-fold cross-validation [2], so the two cannot be compared directly. Published systems also use several further steps that we left out of this baseline on purpose: more views per candidate [3], 3D networks [22], many more negative samples, and longer training. Each of these is a clear way to strengthen the CT branch. The pipeline built here (geometry-aware cropping, patient-level splitting, shared training settings, calibration and Grad-CAM) is reused unchanged in Phases 2 and 3. [add what the Grad-CAM maps show: do they sit on the nodule?]

6.4 Grad-CAM

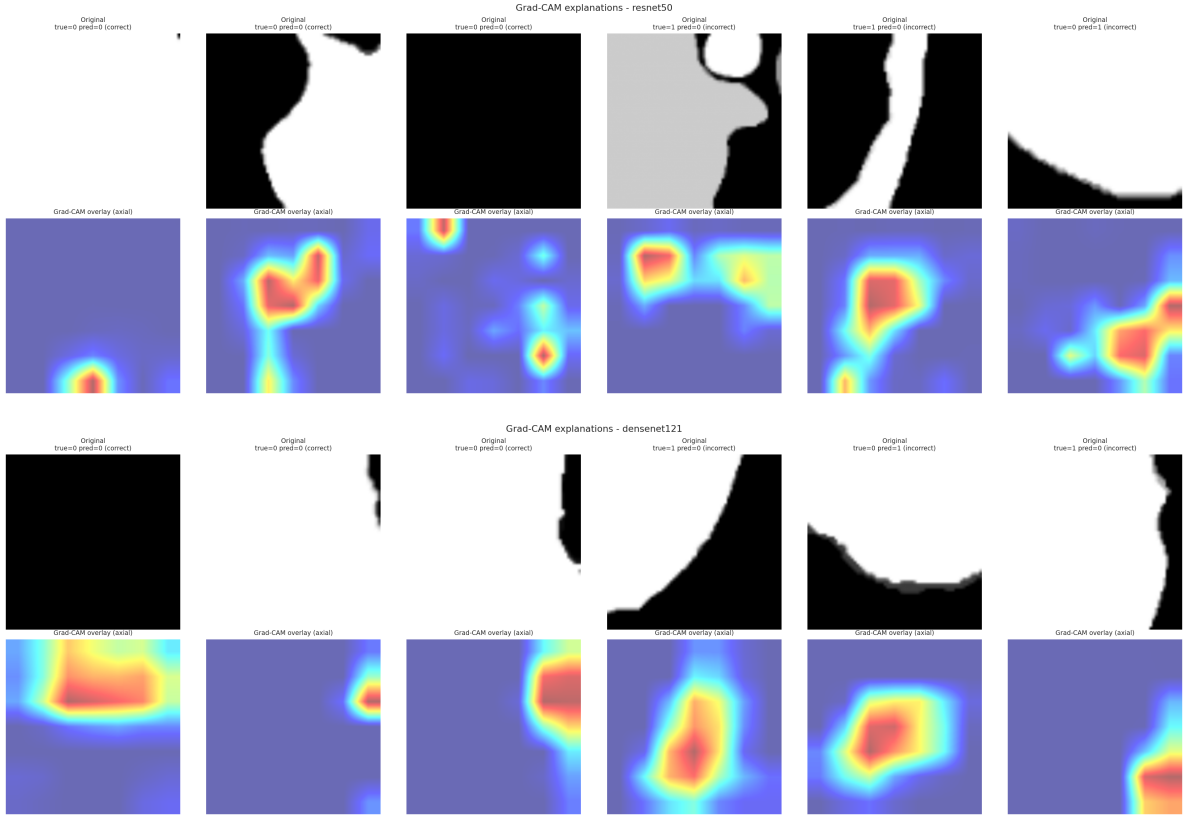


Figure 9: Grad-CAM maps (axial view) for ResNet50 (top) and DenseNet121 (bottom): three correct and three wrong test patches each. The other two models are in Appendix A.

7 Approaches we did not use in Phase 1, and why

- **Vision Transformers.** ViTs have little built-in preference for local image structure and need very large pretraining sets to work well [21]. With a few thousand small patches, a CNN is the safer choice. Attention is used later, in the Phase 2 fusion model, where it combines CT and PET features rather than replacing the CNN.
- **Full 3D CNNs.** 3D networks such as that of Dou et al. [22] use the whole nodule volume and do well on LUNA16. However, they need much more GPU memory, and ImageNet weights cannot be used directly in 3D. The three-view input gives us much of the 3D shape information at 2D cost. A 3D model is a possible later extension.

- **Training from scratch.** Too few images; see Shin et al. [6] and Tajbakhsh et al. [7].
- **Ensembles of one model family** (for example ResNet18/34/50). Such models tend to make the same mistakes, so voting helps less [16].

8 Limitations

- We use one fixed 70/15/15 split. The official LUNA16 protocol is 10-fold cross-validation over the ten subsets, and it reports FROC/CPM at the scan level. Our numbers are therefore candidate-level and cannot be compared directly with the challenge leaderboard.
- Negatives are randomly sampled at 1:3, so precision here is higher than it would be on the full candidate list.
- The label is nodule vs. non-nodule, not malignancy.
- Three orthogonal planes do not capture the full 3D shape of irregular nodules.
- Results come from one test split without confidence intervals, so differences of a few hundredths between models should be read with care. Cross-validation with bootstrap confidence intervals is used in Phases 2 and 3.

A Per-model reports and remaining Grad-CAM maps

Each report shows train/validation loss and accuracy, validation ROC-AUC and PR-AUC per epoch, and the test-set ROC curve, PR curve and confusion matrix.

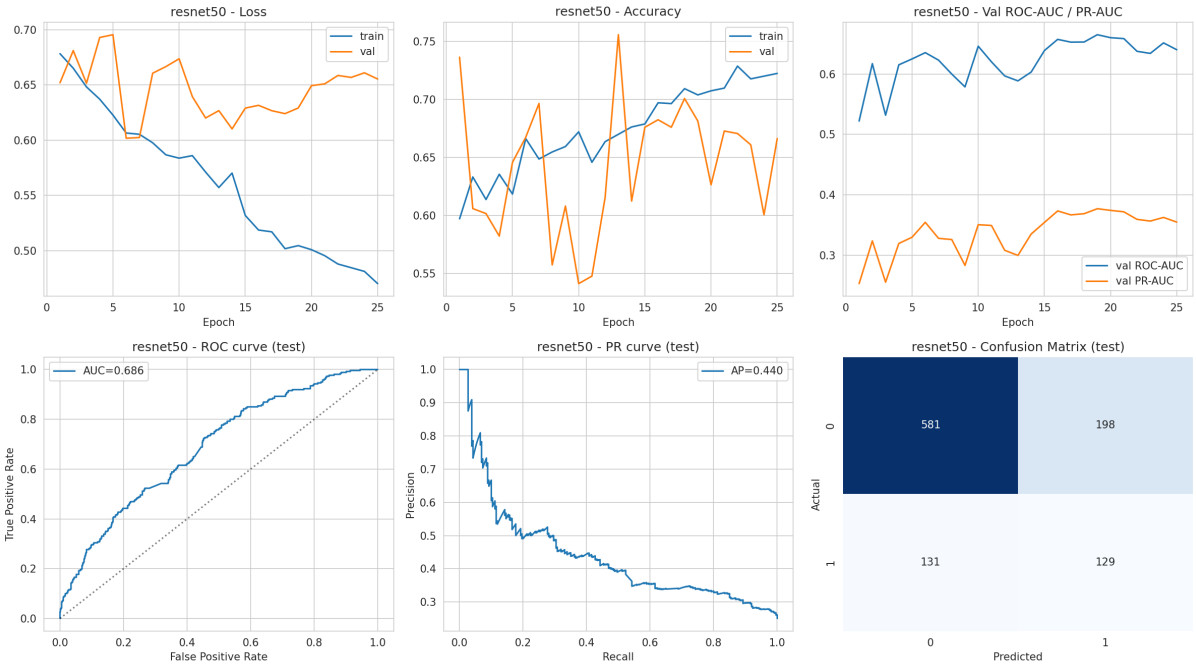


Figure 10: ResNet50.

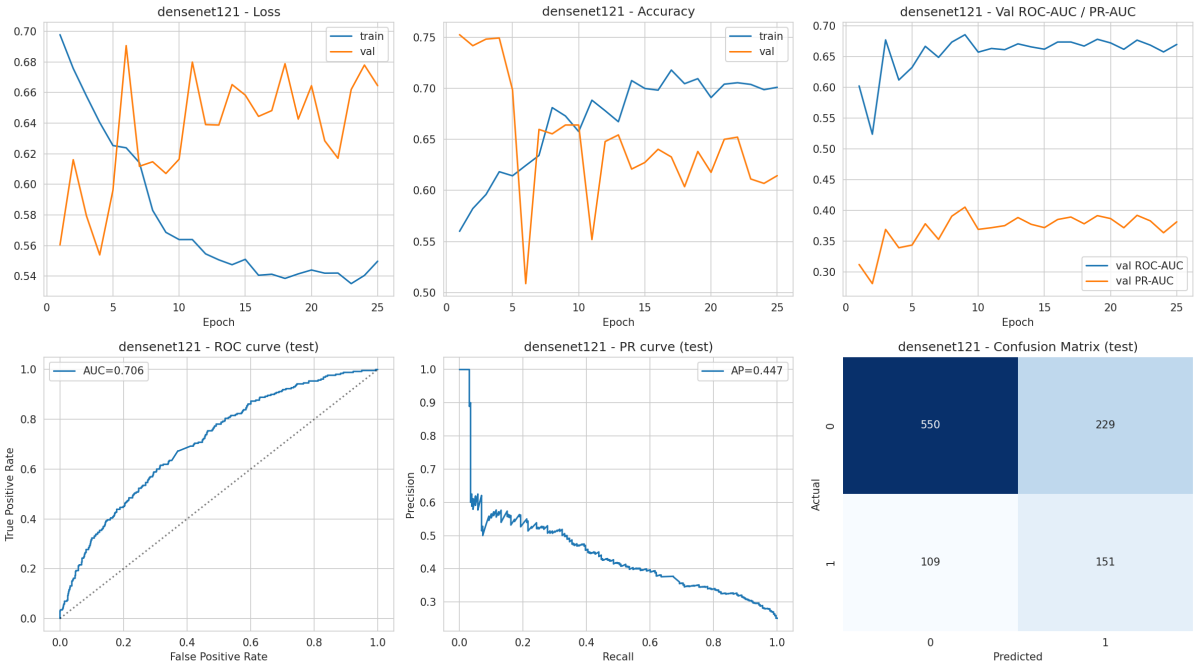


Figure 11: DenseNet121.

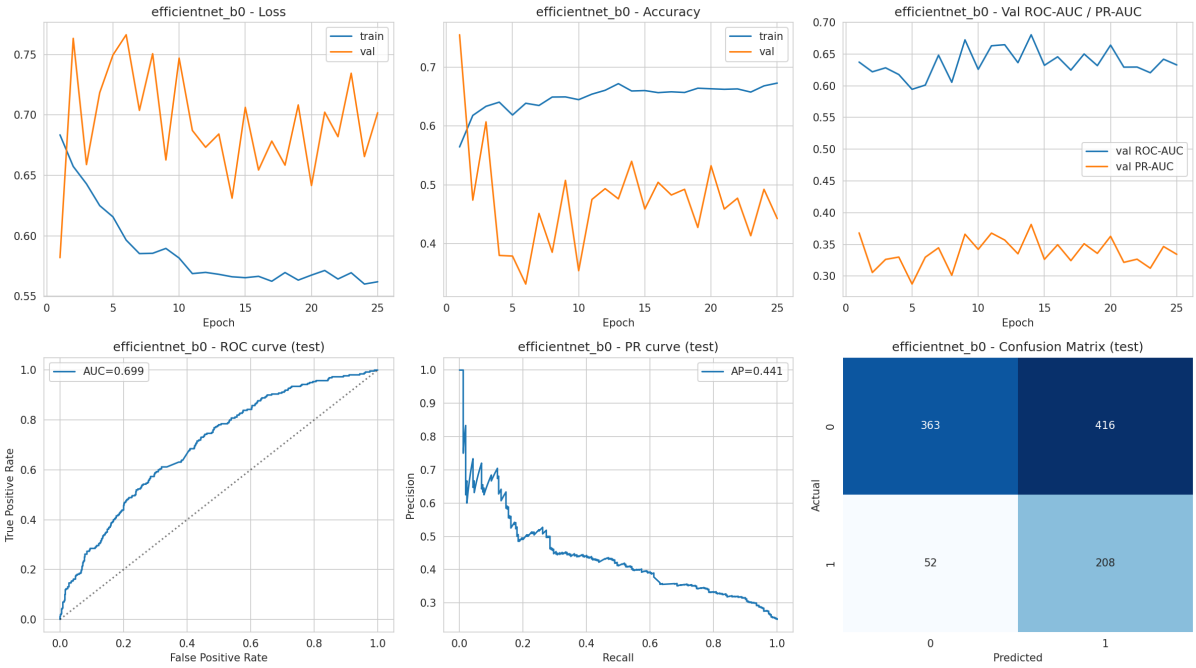


Figure 12: EfficientNet-B0.

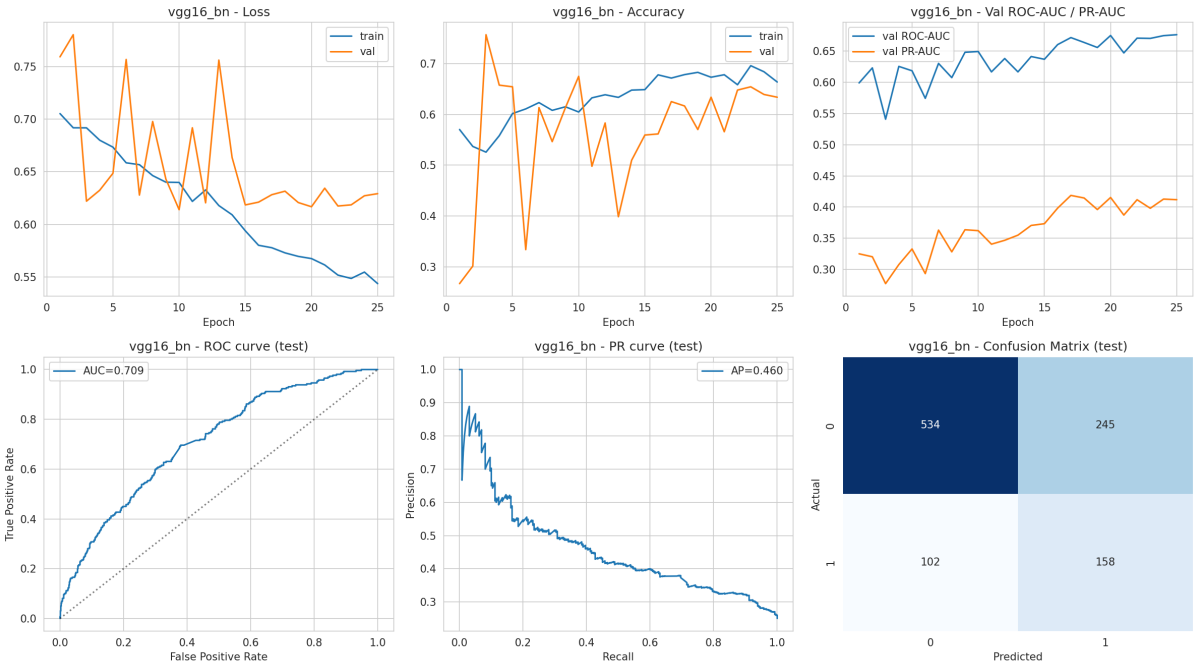


Figure 13: VGG16-BN.

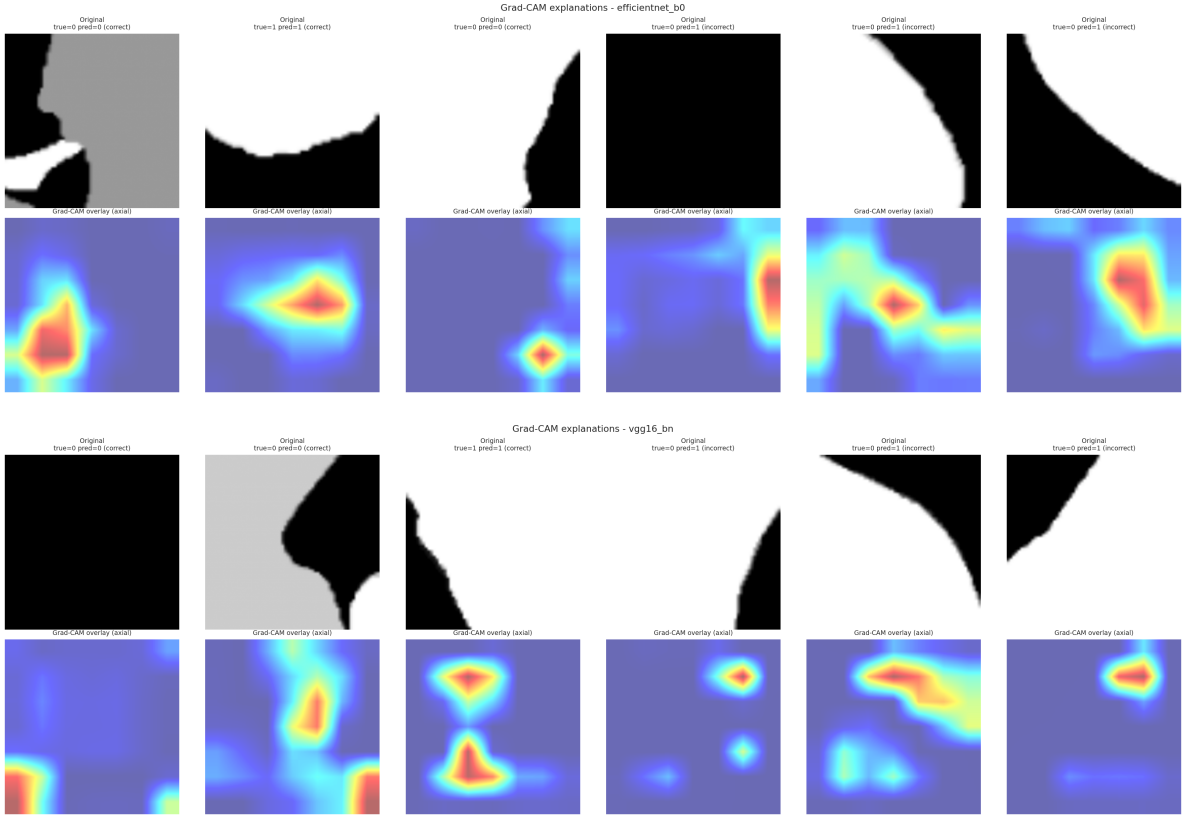


Figure 14: Grad-CAM maps for EfficientNet-B0 (top) and VGG16-BN (bottom).

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